USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 24 LAB - Null Steering
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.9: I can calculate null-steering weights and implement pattern nulls on the ADALM-PHASER.

This packet is the L24 Lab, in three parts, and you turn it in once, complete, before Lesson 25. **Part 1, Predict**, is yours to do before the lab period: the shaded Predicted cells of each table, what it should read on your kit, and the gains and phases you type in for Procedure A. **Part 2, Measure**, is your own kit's readings in the white Measured cells of the same tables, recorded in class. **Part 3, Analyze**, is yours to write after the lab.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt. I grade each Part 3 answer on four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before the lab period)

1. **Predict.** Fill in the gains and phases below, and every shaded **Predicted** cell of Tables A to D. Work at the HB100 frequency your kit reported with **Find HB100** in an earlier lab. If you did not record one, use the nominal 10.525 GHz and say so. Lesson 24's Pre-Lab frame names the frame that works each method, at 10.0 GHz. Attach your work for one row of each table on your own paper; for the gains and phases, your script is enough.

   - **The array.** $N = 8$ patches, E1 to E8, on a $d = 14$ mm pitch, with $+\theta$ on the E8 side. Use $c = 3.00 \times 10^8$ m/s.
   - **The hardware.** Each ADAR1000 sums four patches, so the two digital channels are E1 to E4 and E5 to E8. Its phase shifters step in 2.8125° and its gains in 1% of full scale, and the Element Gains sliders are linear in voltage.
   - **Procedure A.** The beam is held at broadside and the null is at $\theta_1 = +22.5^\circ$.
   - **Procedure C.** The HB100 is the target, at boresight. A simulated interferer arrives 10 dB stronger, first at $+20^\circ$ and then at $+30^\circ$.
   - **Levels.** Report them as the tables do: Tables A and B relative to the frozen uniform peak, and Peak Gain in Table C as its change from Manual's.

   Your frequency, and the gains and phases you type in at Steps 7 and 8:

Frequency								
Steps 7, 8: element	1	2	3	4	5	6	7	8
Gain (%)								
Phase (°)								

Part 2: Measure (your kit, in class)

Follow the procedure on the L24 Lab page, and fill in every white **Measured** cell as you take each reading.

Freq = | Rx Gain = | Distance =

2. **Table A: the static notch** (Procedure A, Steps 6–10). Freeze the uniform trace first. If its two first sidelobes differ by more than about 2 dB , recalibrate before you go on. Read peaks with **Peak Gain** and angles with **Angle** set to **Peak**.

Uniform SL nearest $+22.5^\circ$ (dBc)		**Null-steered peak (dB)**		**Notch bottom (dB)**		**Notch angle**	
Predicted	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**

3. **Table B: the difference null** (Procedure B, Steps 11–15). Beam 1 Phase at 180°, Mode Manual, uniform taper.

Null depth (dB)		**Null angle**		**Left peak angle**		**Right peak angle**	
Predicted	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**

4. **Table C: MVDR against an interferer** (Procedure C, Steps 16–22). Simulated interferer at $+20^\circ$, Power 0 dBFS, about 10 dB above the target; target your HB100 at boresight; beam held at 0° in Signal vs Time; MVDR with $K = 1024$ snapshots. Each measured entry is the median of ten readings. Predict Peak Gain as its change from Manual's.

$+20^\circ$	**Interferer vs Target**		**Peak Gain**		**MVDR Suppression**
	Predicted	**Measured**	**Predicted**	**Measured**	**Measured**
Manual			reference		
MVDR, load 0.1	–				

Rectangular, held	**Null angle**		**Conventional at null**		**Null depth**
	Predicted	**Measured**	**Predicted**	**Measured**	**Measured**
MVDR, load 0.1					

$+30^\circ$**, Peak Gain**	**MVDR, load** 0.001		**MVDR, load** 0.1		**Manual**
	Predicted	**Measured**	**Predicted**	**Measured**	**Measured**

5. **Table D: monopulse** (Procedure D, Steps 23–27). The Tracking preset, Manual beamformer, both digital beams at 0°, uniform taper.

Delta null depth (dBc)		**Delta null angle**		**Error zero crossing**		**Error sign below, above**	
Predicted	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**

Stand's scale		-20°	-10°	$+10^\circ$	$+20^\circ$
Tracked angle	**Predicted**				
	Measured				
How far it follows	**Predicted**				
	Measured				

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions.

6. **Three nulls.** Compare your notch (Table A), your difference null (Table B) and your delta null (Table D) with each other and with your predictions. Explain what set the depth and the angle of each one on your kit, and point to the readings that support each explanation.

7. **MVDR at two angles.** Compare your Manual and MVDR readings with the interferer at $+20^\circ$ and at $+30^\circ$ (Table C), and with your predictions. Explain what MVDR did at each angle. The interferer is simulated: say which of your conclusions would hold for a real interferer in this room, and what Table C doesn't explain about one.

8. **The tracker.** Compare your Table D readings with your predictions: the beam sweep of Steps 25 and 26, and the tracked angles of Step 27 against the stand's scale. How closely did the loop point the beam at the HB100, and what set that accuracy? From what you saw as you moved the HB100, say what limits how fast a source this loop can follow.

Documentation: _____